Skills Summary

Finding Unknown Factors with Related Multiplication Facts

Use this reference while completing your worksheet or when studying.

The one big idea: every division question is a multiplication question with a factor missing. "Total, number of groups, size of group" — know any two, find the third.

1. Unknown factor from an array (unknown-factor-from-array)

Model:

rows $\times$ (how many in each row) $=$ total.

Cover one of the three numbers and it becomes the unknown factor. If you know the total and the number of rows, ask: *how many rows of what make the total?*

Example: An array has 3 equal rows and 18 counters in all. How many counters are in each row?
Step 1. The number of rows and the total are known, so the missing number is a factor: write $3\times ? = 18$.
Step 2. Count up by threes until you reach the total: $3, 6, 9, 12, 15, 18$ — that took six threes. Each row holds **6** counters.

Try it: An array has 5 equal rows and 40 counters in all. How many counters are in each row?

check: **8**

2. Unknown factor in an equation (unknown-factor-equation)

Rule:

To solve $a\times ? = c$ (or $? \times a = c$), read it as *"a times what gives c?"* Order does not matter in multiplication, so a blank in front is solved exactly the same way. Skip-count by a, or recall the times-table row for a.

Example: Find the unknown factor: $8\times ? = 40$.
Step 1. One factor and the product are known, so this is an unknown-factor question: eight times what gives 40?
Step 2. Skip-count by eights: $8, 16, 24, 32, 40$ — five eights. The unknown factor is **5**.

Try it: Find the unknown factor: $? \times 4 = 32$.

check: **8**

3. Equal-group stories (unknown-factor-word-problem)

How to set it up:

Find the **total** and the **number of equal groups** in the story, then write (number of groups) $\times$ (size of each group) $=$ total, leaving the unknown blank. Words like *shares equally, each, per, in all* tell you which number is which.

Example: 24 apples are packed equally into 3 baskets. How many apples are in each basket?
Step 1. Three equal groups with a total of 24, and the group size is missing — so write $3\times ? = 24$.
Step 2. Three of what makes 24? Count by threes to 24, or use the fact $3 \times 8 = 24$. Each basket holds **8** apples.

Try it: 35 marbles are packed equally into 5 bags. How many marbles are in each bag?

check: 7

4. Using a related division fact (related-division-fact)

Fact family:

One trio of numbers makes four facts. From $4 \times 9 = 36$ you also get $9 \times 4 = 36$, $36 \div 4 = 9$, $36 \div 9 = 4$.

So a division you have never practised is just a multiplication fact you already know, read backwards.

Example: You already know $9 \times 4 = 36$. Use it to find $36 \div 9$.
Step 1. Dividing by 9 asks for the missing factor in $9\times ? = 36$, and the known fact already fills that blank.
Step 2. The fact says nine fours make 36, so the missing factor is the other number in the fact. Therefore $36 \div 9 = \mathbf{4}$.

Try it: You know $7 \times 8 = 56$. Use it to find $56 \div 8$.

check: 7

(!) Watch out: The blank is a *factor*, never the total. In $6\times ? = 54$ the answer is not 54 and not 6 — it is the number of sixes it takes to reach 54. And in a two-step story, answer the question that was asked: the mystery number may only be step one.